

TEACH FOR AMERICA: Addressing the Recruitment Challenge



Presented By: Team 5

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TEACH FOR AMERICA

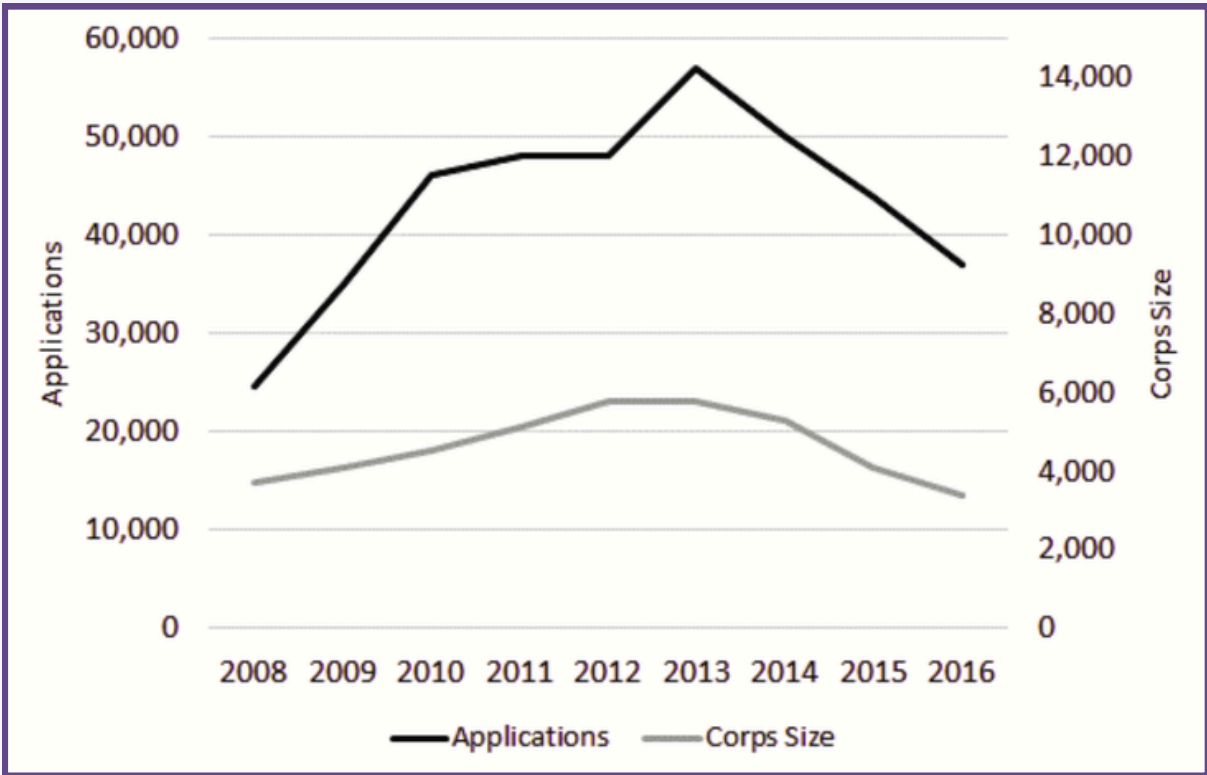
A nonprofit organization that recruits and trains college graduates and professionals to teach in under-resourced schools across the United States.

Limited Staff & Resources



Limited staff resources were strained by spending time on applicants who ultimately failed to complete the application process.

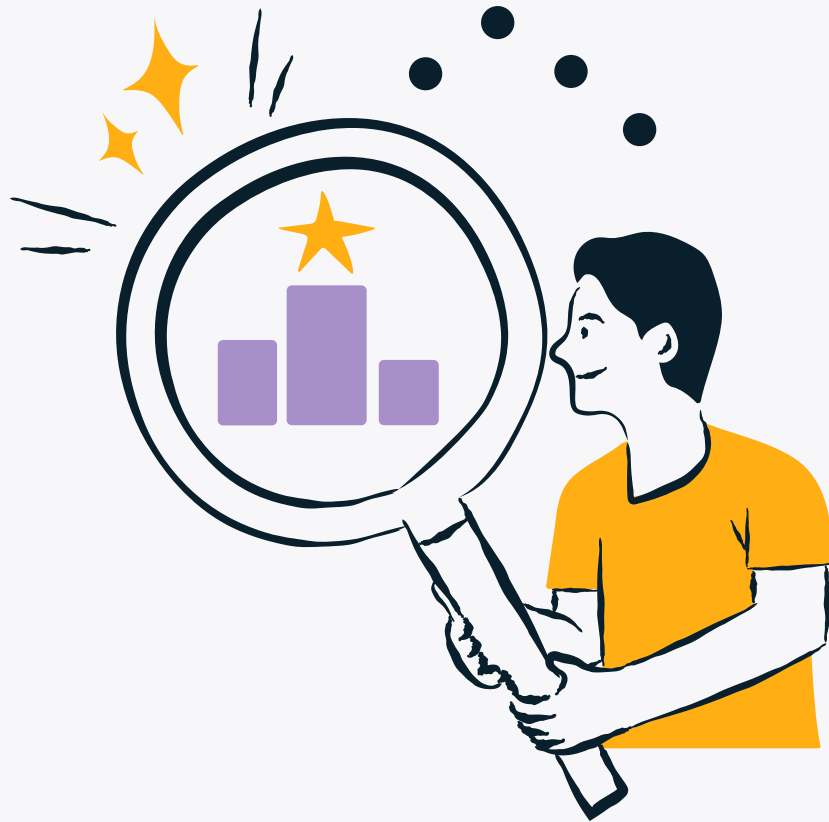
Declining Applications



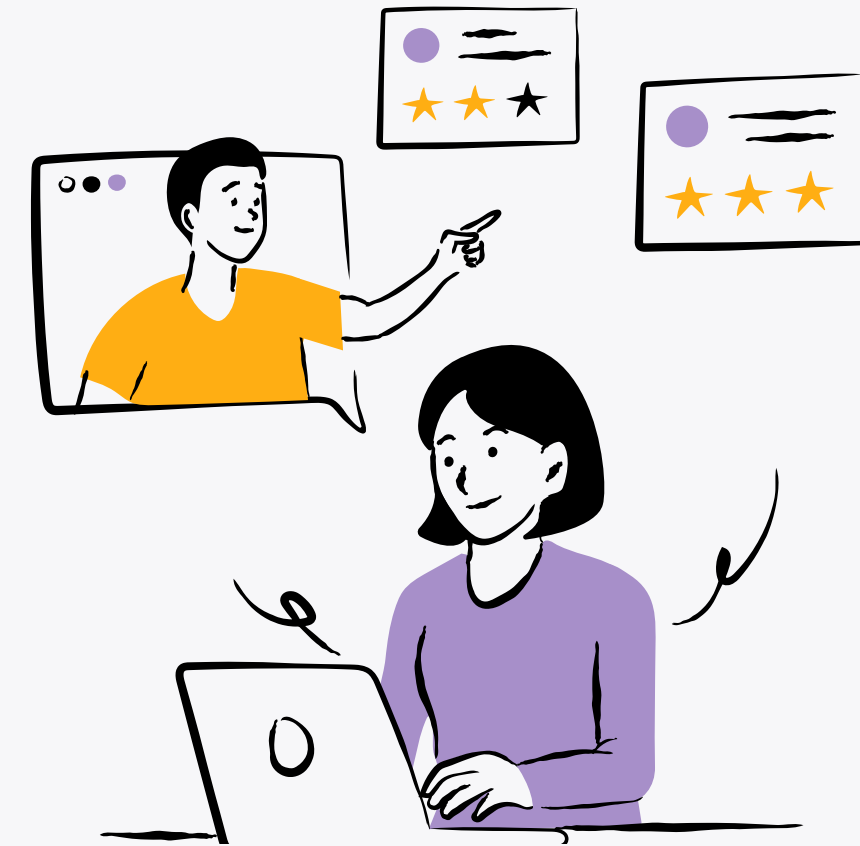
Teach For America's Application and Corps Size Totals, 2008–2016

- Applications peaked at about 57,000 in 2013
- Declined by 12% in 2014
- 10% in 2015
- 16% in 2016

OUR GOALS

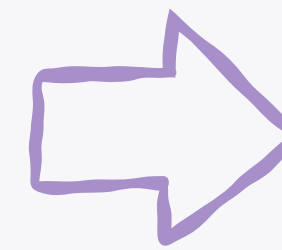
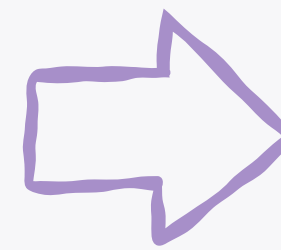
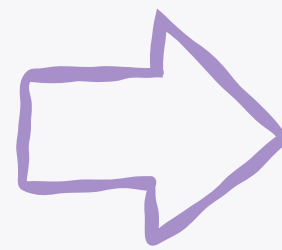
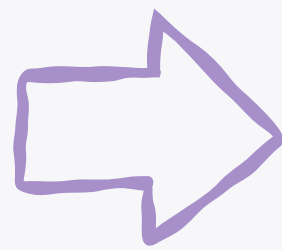


Build a model to identify high-potential applicants to efficiently allocate Recruiter resources.



Streamline the recruitment process to enhance user experience.

CURRENT RECRUITMENT PROCESS



Online Application

(GPA, Resume, Personal Statement)

Document Submission

(References, Transcript)

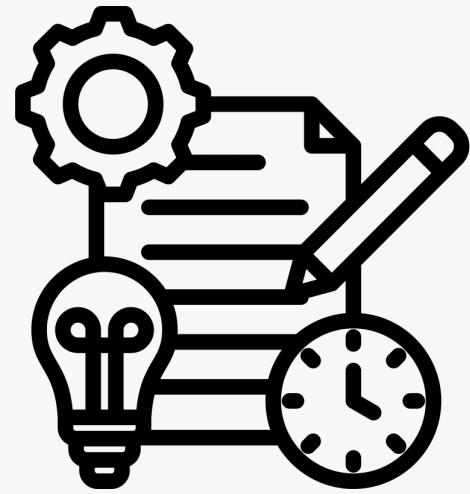
In-Person Interview & Demo Lesson

Selection & Region Placement

Offer Acceptance & Training

- **Excessive Upfront Effort:** Long application —> high abandonment
- **High Uncertainty/Low Confidence:** High investment, low confidence
- **Costly Requests Too Early:** Transcripts/references requested too soon
- **Delayed Human Interaction:** Late recruiter contact
- **Protracted Timeline:** Long 4-6 week process

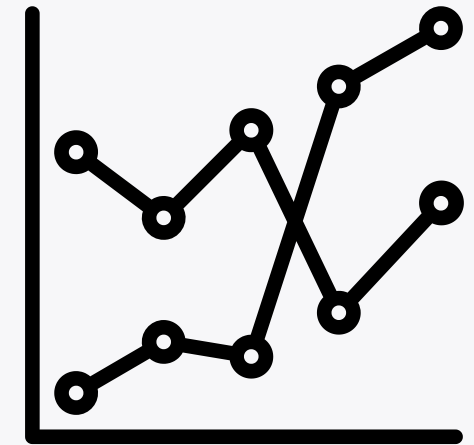
DATASET CHALLENGES



Inconsistent data
dictionary

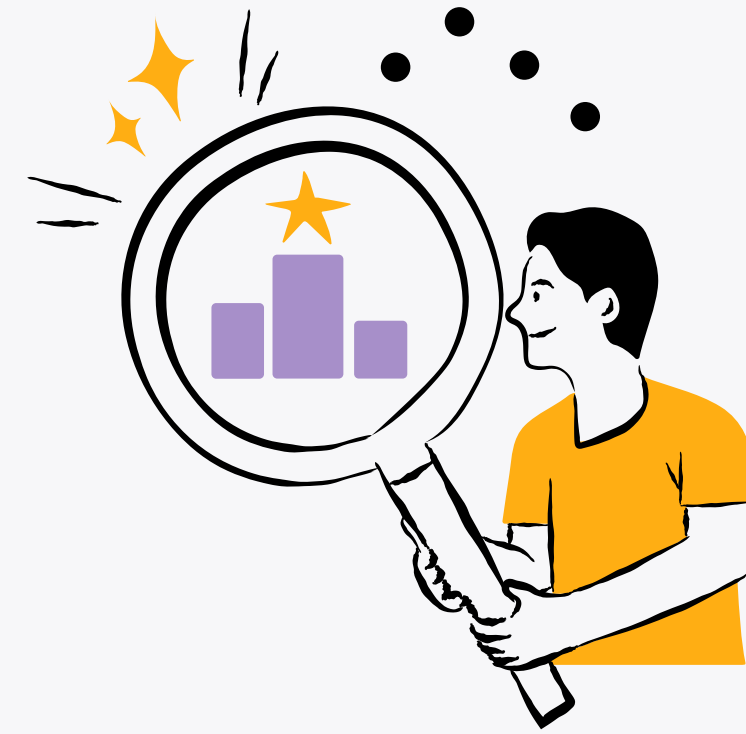


Missing & noisy
values



Variables with
low predictive
power

OUR APPROACH



• Cleaning

- X 1. ~~Admit Year~~
- X 2. ~~Math/Sci/STEM~~
- 3. GPA
- X 4. ~~Major (cont)~~
- X 5. ~~Minor (cont)~~
- 6. School Selectivity (~~cont~~) → 0-6
- X 7. ~~Region (1-5)~~
- 8. Sign up date - date applicant registered an account
- 9. Started date - date started application process TFA
- 10. app deadline
- 11. Submitted date - date submitted app. process TFA
- 12. attended event (0/1)
- 13. essay length 1, 2, 3 / 100
- X 14. ~~Essay Sentiment~~

Outcome: Completed Ad. Process

• Final

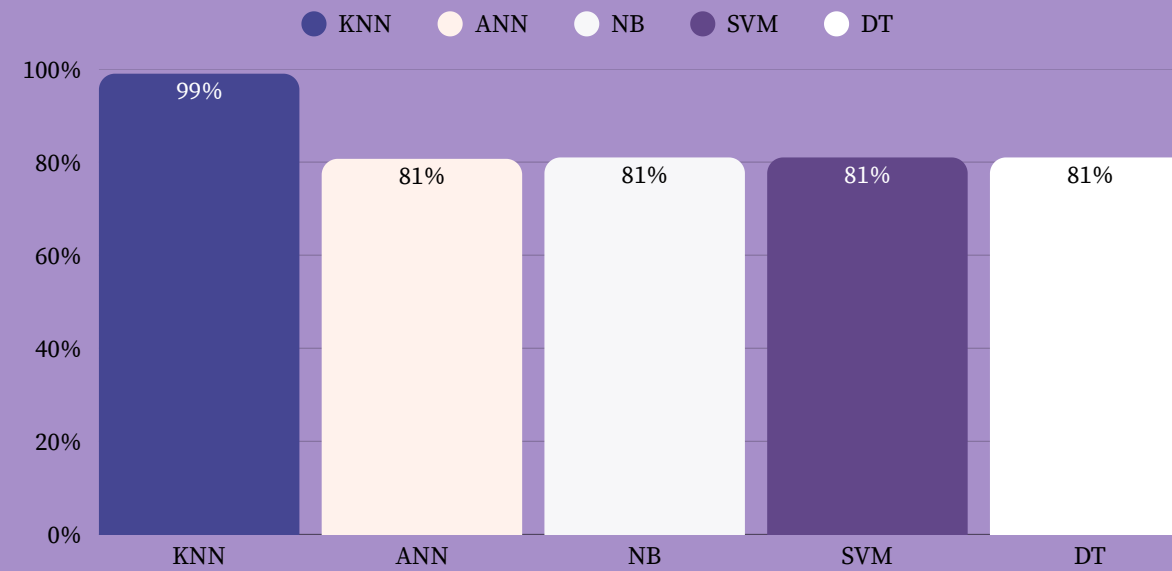
1. GPA
2. School Selectivity (0 - unknown, 6 - most likely)
3. Days Sign to start ← start date - sign up date
4. Days Start to submit ← Submit date - start date
5. Days to submit deadline ← app deadline - submit date
6. Attended event
7. Essay length 1, 2 & 3 / 100

9 variables

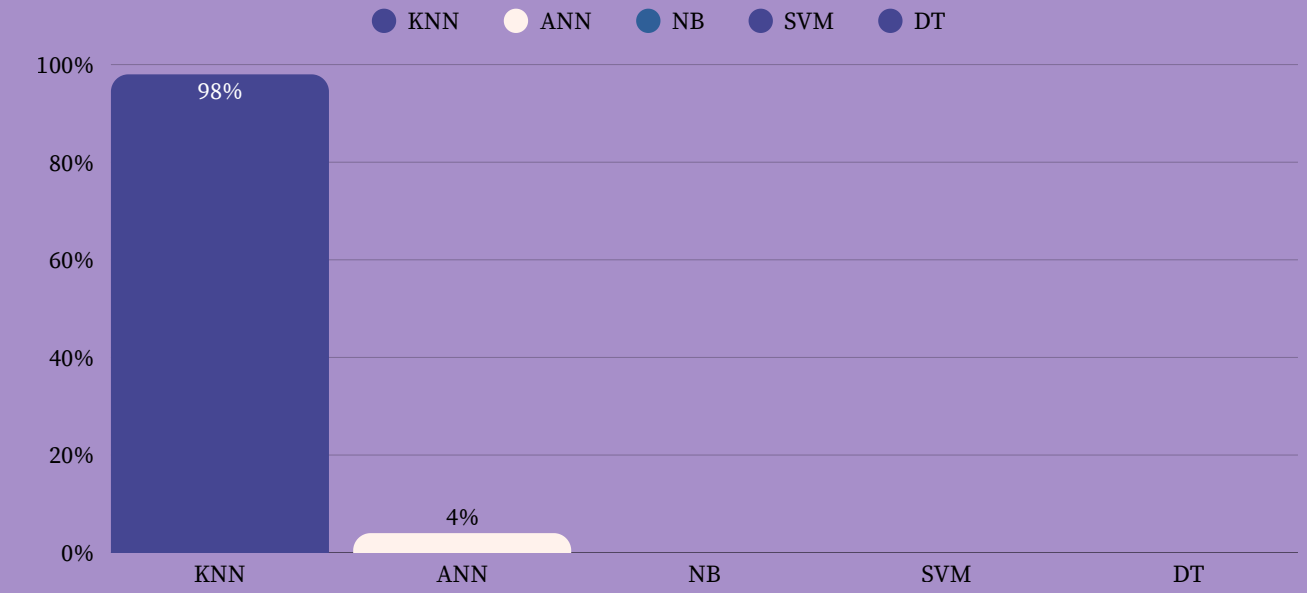
Outcome: Completed app process

MODEL EVALUATION

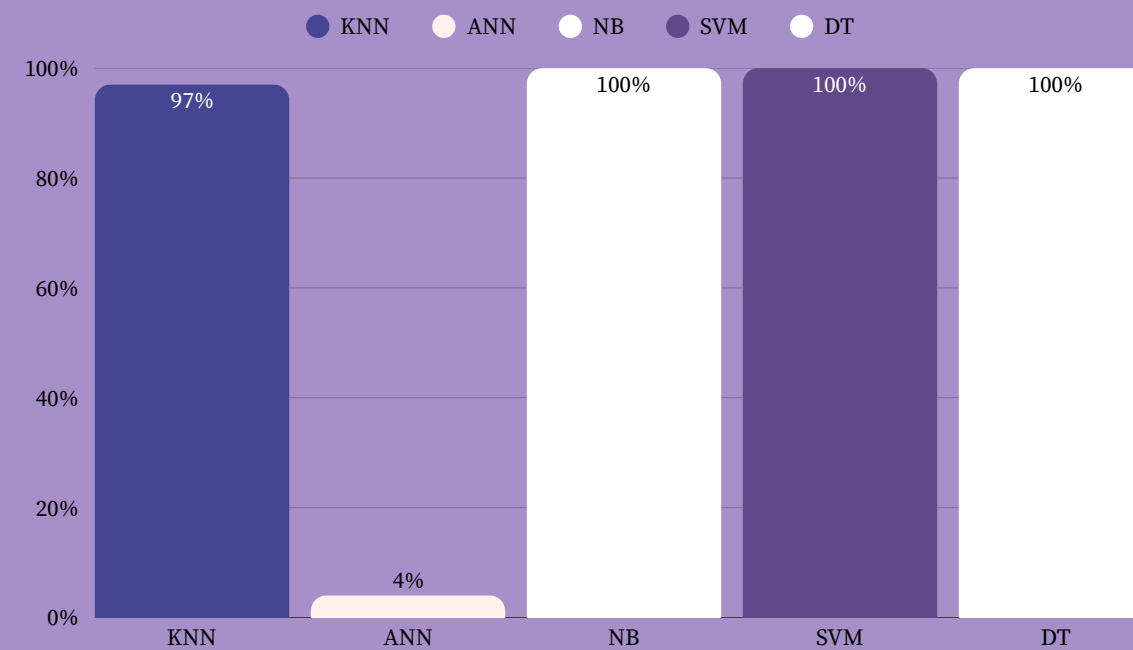
Accuracy Values by Model



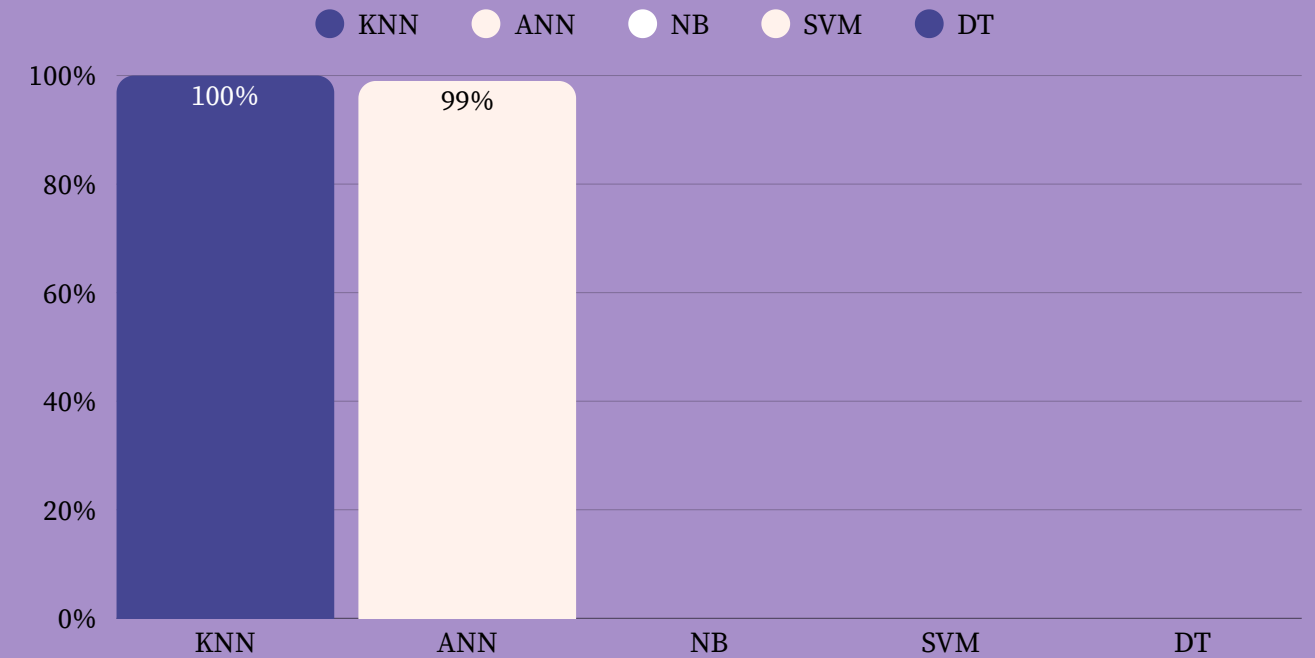
Kappa Values by Model



Sensitivity Value by Model



Specificity Values by Model



We chose KNN as the best model, since KNN has the highest accuracy score, the highest kappa scored and also the best split of sensitvty and specificty scores, as seen in the graphs above.

NEW APPLICANT PREDICTIONS

Meet Jim Johnson, an upcoming graduate of Stanford University. He has recently learned about the Teach for America program at a recruiting event and is interested in applying.

Using the model previously discussed we will predict whether or not he completes the admissions process based on the information provided from his fictional application:

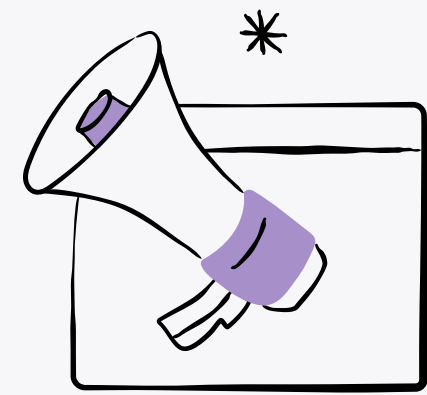
Cumulative GPA: 3.4	Essay 3 Length: 400	Days Start to Submit: 5
Essay 1 Length: 250	Attended Event: 1	Days Submit to Deadline: 10
Essay 2 Length: 25	Days Signup to Start: 2	School Selectivity: 5

We predict he will complete the Admissions Process!

```
> pred_new <- knn(train = train_predictors,  
+                 test  = new.student,  
+                 cl    = train_labels,  
+                 k = 3)  
> pred_new  
[1] 1  
Levels: 0 1
```

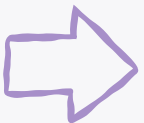
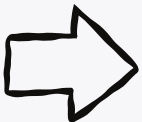


OLD RECRUITMENT MODEL



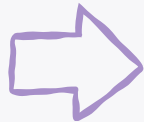
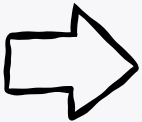
NEW RECRUITMENT MODEL

Online Application
(Academic Information, Resume,
Essay & Personal Statement)



Online Application
(Academic Information, Resume,
Essay & Personal Statement)

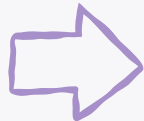
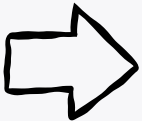
Document Submission
(References & Transcript)



Follow-up Call/Email
(For incomplete applications)

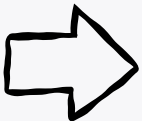
-----Application Process Complete

**In-Person Interview
& Demo Lesson**



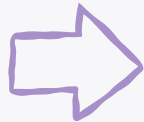
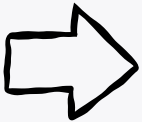
**In-Person Interview
& Demo Lesson**

**Selection &
Region Placement**



Conditional Offer
(References, Transcripts, Teaching
Subject & Region Preference)

**Offer Acceptance
& Training**



**Offer Acceptance,
Training & Placements**

-----Admission Process Complete

BUSINESS IMPACT

High-quality Data



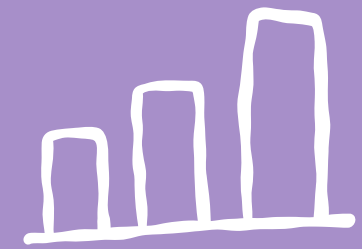
Captures necessary data while drastically reducing friction in the early stages of application.

Improved Allocation



Allows TFA associates to better allocate time and resources based on seriousness data.

More Applications



The new model encourages application completion and is confident to attract and retain higher-quality candidates.



THANK YOU